

Home-Foreign Elasticity Schedules and the Gains from Trade

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- Answer depends on the domestic expenditure share and a trade elasticity that is constant
- Same trade elasticity governs substitution btw home-foreign and across foreign sources
- Elasticity estimated from expenditure reallocation across foreign sources

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Theoretical Contributions:

- 1 GFT depend on one-dimensional schedule of home-foreign elasticities as domestic share $\uparrow$
 - ▶ GFT independent of substitution across foreign sources
 - ▶ Proven w/ general homothetic prefs on a path on which all foreign prices $\uparrow$ proportionally
- 2 Price path feasible in GE: schedule can be estimated directly via tariff variation
- 3 Identification argument assuming common sector-specific schedule across countries

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Empirical Contributions:

- Directly estimate sector-level elasticity schedules from changes in home expenditure relative to aggregate foreign expenditure across destinations w/in sector
- Qualitative result: Elasticity schedules are increasing in domestic expenditure shares
- Example quantitative results: w/in sector, moving from 25th to 75th percentiles
 - ▶ OECD ICIO: From 4.01 to 5.88
 - ▶ GTAP: From 2.55 to 3.48
- Sensitivity: Quantitatively stable across extensive sensitivity, including two IV strategies

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Quantitative Contributions:

- Combine estimates + assumptions on sectoral aggregation and IO to measure GFT
- e.g., OECD: Mean 12.82 log pts; compare to 15.60 if each elasticity fixed at its initial level
- New Mechanism: A lower initial domestic expenditure $\Rightarrow$ a lower effective elasticity

Literature

- GFT sufficient statistics: ACR (2012), CR (2018), and Ansari et al. (2026)...
 - Directly estimate welfare-relevant elasticity schedule
 - Provide generalized quantitative Armington, Krugman, Melitz gravity models
-
- Nonparametric trade counterfactuals: Adao, Costinot, and Donaldson (2017); Adao, Arkolakis, and Ganapati (2026)
 - Identify the low-dimensional home-foreign factor demand system using tariff variation
-
- International macroeconomics: Backus et al. (1994), Heathcote and Perri (2002), Corsetti et al. (2008), Feenstra et al. (2018)...
 - Relative to Feenstra et al. (2018), allow the home-foreign elasticity to vary with the domestic expenditure share and estimate its schedule across countries and sectors

Gains from Trade

Environment
Result given a price path
Feasibility of that path

Environment

Armington production

- Many countries $i \in \mathcal{N}$ endowed with L_i units of labor, linear production $y_i = a_i L_i$
- Each country a monopolist over its own variety, balanced trade, competitive markets
- Let w_i denote the wage and $Y_i = w_i L_i$ nominal income
- Unit production cost $c_i = w_i / a_i$, iceberg trade cost $\mathcal{D}_{ki} \geq 1$, so $p_{ki} = c_i \mathcal{D}_{ki}$

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+ otherwise general homothetic preferences

- Country n has prefs over $x_n = (x_{ni})_{i \in \mathcal{N}}$ represented by homothetic utility function u_n
- Unit expenditure function $e_n(p_n)$ so real income is $W_n \equiv Y_n / e_n(p_n)$

Moving Along a Path from Trade Toward Autarky

- Start from the trade equilibrium, indexed by $m = 1$
- At each m , multiply relative foreign prices by $m \geq 1$

$$\pi_{ni}(m) \equiv \frac{p_{ni}(m)}{p_{nn}(m)} = \begin{cases} 1 & \text{if } i = n, \\ m \frac{p_{ni}(1)}{p_{nn}(1)} & \text{if } i \neq n. \end{cases}$$

- Relative prices in foreign bundle are fixed, so foreign goods form a Hicks composite

The Welfare Effect of a Small Move Along the Ray

- $H_n(m) \equiv e_n(\pi_n(m))$: unit expenditure measured in units of the home good
- Common $\uparrow$ in foreign prices $\uparrow$ expend. in prop. to foreign expenditure share

$$\frac{d \ln H_n(m)}{d \ln m} = \sum_{i \neq n} \lambda_{ni}(m) = 1 - \lambda_{nn}(m)$$

share of n 's expenditure from i



which used Shephard's lemma (and every foreign price changes by m)

- Combined with $W_n(m) = \frac{Y_n(m)}{e_n(\mathbf{p}_n(m))} = \frac{p_{nn}(m)a_n L_n}{p_{nn}(m)H_n(m)} = \frac{a_n L_n}{H_n(m)}$, we have

$$d \ln W_n(m) = -(1 - \lambda_{nn}(m)) d \ln m$$

$\Rightarrow$ Welfare loss from small foreign price increase doesn't depend how foreign expenditure is allocated

Gains from Trade

Convert $d \ln m$ to $d \ln \lambda_{nn}$ using home-foreign elasticity

$$\varepsilon_n(m) \equiv \frac{d \ln [X_{nn}(m)/X_{nF}(m)]}{d \ln m} \quad \Rightarrow \quad d \ln m = \frac{1}{1 - \lambda_{nn}} \frac{d \ln \lambda_{nn}}{\varepsilon_n(m)}$$

For presentation, assume $\varepsilon_n(m) > 0$: $\lambda_{nn}(m)$ rises strictly with m , so write $\varepsilon_n(\lambda_{nn})$

For finite gains, assume home good alone can deliver one unit of utility:

$$d \ln W_n = - \frac{d \ln \lambda_{nn}}{\varepsilon_n(\lambda_{nn})} \quad \Rightarrow \quad \ln \frac{W_n^T}{W_n^A} = \int_{\lambda_{nn}^T}^1 \frac{d \ln \lambda}{\varepsilon_n(\lambda)}$$

Sufficient stats: initial share (λ_{nn}^T) + home-for. elasticity schedule btw ($\lambda_{nn}^T, 1$)

Substitution across foreign sources is irrelevant

also true if allow $\varepsilon_n(m) \leq 0$

What is special about the proportional price path?

Along a path Γ from trade ($t = 1$) to autarky, define local log relative (i to n) price growth:

$$\alpha_{ni}(t) \equiv \frac{d \ln \pi_{ni}(t)}{dt} \quad (i \neq n) \quad \text{Baseline: } \alpha_{ni}(t) = \frac{d \ln m(t)}{dt}$$

With $\mu_{ni}(t) \equiv X_{ni}(t)/X_{nF}(t)$, define cumulative log relative foreign-home price change:

$$z_n(t) \equiv \int_1^t \sum_{i \neq n} \mu_{ni}(u) \alpha_{ni}(u) du \quad \text{Baseline: } z_n(t) = \ln m(t)$$

The home-foreign elasticity along this path is

$$\varepsilon_n^\Gamma(t) \equiv \frac{d \ln [X_{nn}(t)/X_{nF}(t)]/dt}{z'_n(t)}$$

With $z'_n(t) > 0$ and $\varepsilon_n^\Gamma(t) > 0$, the previous GFT formula applies using $\varepsilon_n^\Gamma(\lambda)$

Details

- General path: constructing $z_n(t)$ requires foreign shares along the path
- Baseline: all $\alpha_{ni}(t)$ coincide, so these shares drop out

Implementing the Path in GE

- One-dimensional schedule $\varepsilon_n(\lambda)$ is sufficient to measure GFT
- Can trade-policy variation generate this price ray when wages adjust in GE?

Implementing the Path in GE

- One-dimensional schedule $\varepsilon_n(\lambda)$ is sufficient to measure GFT
- Can trade-policy variation generate this price ray when wages adjust in GE? Yes

Proposition. Fix an initial equilibrium without tariffs at $m = 1$. Hold all primitive parameters fixed. For every finite $m \geq 1$, there exist tariffs or subsidies imposed by country n (with revenue rebated lump sum) such that there is a competitive equilibrium satisfying $p_{ni}(m)/p_{nn}(m) = \pi_{ni}(m)$ for every $i \neq n$. At $m = 1$, the implementing trade taxes can be set to zero. As $m \rightarrow \infty$, country n approaches autarky, where imports and tariff revenue are zero.

Proof outline

- In practice, our data do not trace this ray
- So why establish feasibility?
 - shows that one-dimensional measurement feasible in GE w/ general homothetic preferences
- We next ask what additional restrictions allow us to use the price variation we observe

Weak Separability and Identification

Environment + GFT Result
Nonparametric Identification
Imposing a Polynomial Elasticity Schedule

Weak separability

- Source-specific prices do not move proportionally in the data
- **Restriction: Weak separability btw home and foreign goods**
- Replace Hicks composite w/ a weakly separable foreign aggregate and its exact price index

Many sectors

- **Within sectors:** Weak separability btw home and foreign goods
- **Across sectors:** A general homothetic aggregator combines the sectoral price indexes
- Define the within-sector home-foreign trade elasticity

$$\varepsilon_{ns}(\lambda_{nns}) \equiv \frac{\partial \ln(X_{nns}/X_{nFs})}{\partial \ln(P_{nFs}/p_{nns})}$$

and use the same argument as in the one-sector model to obtain

$$\ln \frac{W_n^T}{W_n^A} = \ln \frac{\mathcal{E}_n((e_{ns}^T \exp(I_{ns}))_{s \in \mathcal{S}})}{\mathcal{E}_n((e_{ns}^T)_{s \in \mathcal{S}})} \quad \text{where} \quad I_{ns} = \int_{\lambda_{nns}^0}^{\lambda_{nns}^1} \frac{d \ln \lambda}{\varepsilon_{ns}(\lambda)}$$

I_{ns} : change in log sectoral unit expenditure, in units of country n 's good in s

- **NB:** Along an aggregate foreign price ray, the aggregate sufficient statistic still applies

Estimating Equation + Restriction

- **Cross-country restriction:** $\varepsilon_{ns}(\lambda) = \varepsilon_s(\lambda)$ within sector s

Estimating Equation + Restriction

- **Cross-country restriction:** $\varepsilon_{ns}(\lambda) = \varepsilon_s(\lambda)$ **within sector** s
- Taking a first-order approx around $\lambda_{nns,0}$ yields

$$\Delta \ln \frac{X_{nns}}{X_{nFs}} = \varepsilon_s(\lambda_{nns,0}) Q_{ns}(\mathbf{C}) + u_{ns}$$

$Q_{ns}(\mathbf{C}) \equiv Q_{ns}^{\mathcal{T}} + Q_{ns}^{\mathcal{C}}(\mathbf{C})$ is the tariff + unit-cost component of relative-price change

$$Q_{ns}^{\mathcal{T}} \equiv \sum_{i \neq n} \mu_{nis,0} \Delta \log(1 + \mathcal{T}_{nis})$$

$$Q_{ns}^{\mathcal{C}}(\mathbf{C}) \equiv \sum_{i \neq n} \mu_{nis,0} (C_i - C_n)$$

u_{ns} : finite-change error, demand shocks, *is* productivity deviations from i 's mean

Parametric Identification via a Polynomial Elasticity Schedule

$$\varepsilon_{s,J}(\lambda) = \sum_{j=0}^J \theta_{sj} \lambda^j$$

- **Any continuous $\varepsilon(\lambda)$ can be approximated arbitrarily well by polynomials**
- **Special cases:**
 - ▶ $J = 0$: Constant-elasticity CES benchmark
 - ▶ $J = 1$: Baseline linear schedule (CDE: Constant Difference of Elasticities)
- **Provide gravity frameworks w/ nested CDE-CES preferences to conduct broader counterfactual analyses:**
Armington, Krugman, Melitz-Pareto gravity models

Estimating Home-Foreign Elasticities

Parametric Specification and Estimating Equation

Baseline linear schedule:

$$\varepsilon_s(\lambda_{nns}) = \theta_{s0} + \theta_{g(s)1} \lambda_{nns}$$

- Sector-specific intercepts θ_{s0}
- Slopes pooled within aggregate $g(s)$

Estimating equation:

$$\Delta \log \frac{X_{nns}}{X_{nFs}} = (\theta_{s0} + \theta_{g(s)1} \lambda_{nns,0}) Q_{ns}(\mathbf{C}) + u_{ns}$$

- One time change removes time-invariant destination-sector demand heterogeneity

Variation Used in Estimation

$$\Delta \log \frac{X_{nns}}{X_{nFs}} = (\theta_{s0} + \theta_{g(s)1} \lambda_{nns,0}) Q_{ns}(\mathbf{C}) + u_{ns}$$

$$Q_{ns}(\mathbf{C}) = \underbrace{\sum_{i \neq n} \mu_{nis,0} \Delta \log(1 + \mathcal{T}_{nis})}_{Q_{ns}^{\mathcal{T}}} + \underbrace{\sum_{i \neq n} \mu_{nis,0} (C_i - C_n)}_{Q_{ns}^C(\mathbf{C})}$$

- **Tariffs:** $Q_{ns}^{\mathcal{T}}$ is more – when foreign expenditure concentrated in i w/ larger tariff $\downarrow$
- **Country costs:** $Q_{ns}^C(\mathbf{C})$ is more – when foreign expenditure concentrated in i w/ larger C_i $\downarrow$
- **Elasticity schedule variation:**
 - ▶ $Q_{ns}(\mathbf{C})$ identifies its level, θ_{s0}
 - ▶ $\lambda_{nns,0} Q_{ns}(\mathbf{C})$ identifies its slope, $\theta_{g(s)1}$
- **Estimator:** NLLS profiles out $\mathbf{C}$ (with $C_{\text{USA}} = 0$); SEs are clustered by destination

Moments

Details

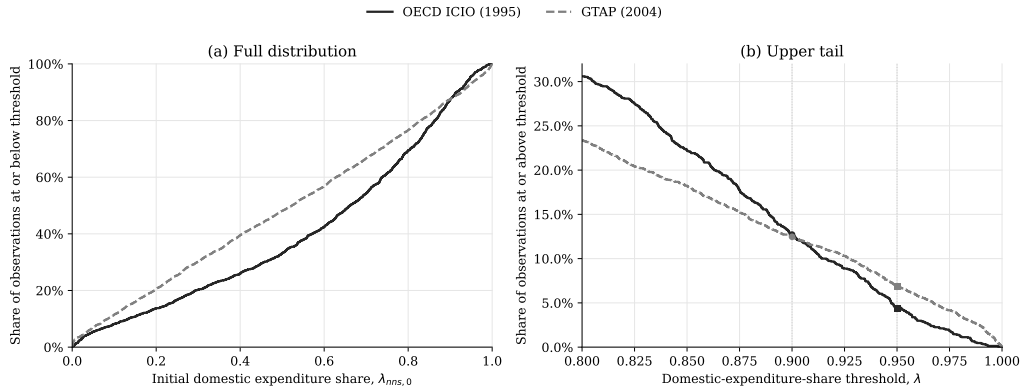
Data and Measurement

Domestic expenditure: Less standard or well measured than international trade flows → use two distinct sources

	OECD ICIO	GTAP
Time change	1995–2007	2004–2014
Destinations	80	120
Manufacturing sectors	19	27
Destination–sector observations	1,510	3,240

- Both measure total absorption, including intermediate and final uses
- Bilateral tariffs come from Teti's Global Tariff Database

Extrapolation: The Distribution of $\lambda_{nns,0}$



- **Observed support:** Both samples contain substantial variation, including domestic expenditure shares near one
- **Extrapolation:** Quantification evaluates each fitted schedule from observed share to one

Estimation Results: OECD ICIO Baseline

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	5.45 (1.88)	4.01 (1.12)	5.12 (1.32)	5.88 (1.49)	0.26
(B)	Consumer manuf.	8.06 (2.44)	2.98 (0.81)	4.67 (1.12)	5.76 (1.39)	0.27
	Producer manuf.	2.05 (2.49)	5.19 (1.56)	5.60 (1.56)	5.89 (1.65)	
(C)	Food, bev., tob.	6.90 (3.91)	3.04 (0.74)	3.73 (0.80)	4.03 (0.87)	0.28
	Textiles/apparel/leather	12.35 (4.09)	0.84 (0.92)	4.19 (1.00)	6.11 (1.48)	
	Wood/paper/printing	-1.32 (4.15)	3.34 (1.04)	3.13 (0.87)	3.01 (0.98)	
	Petrol./chem./pharm./rubber	3.70 (2.80)	4.53 (1.30)	5.18 (1.45)	5.60 (1.61)	
	Minerals/metals	-1.48 (3.14)	7.18 (2.18)	6.90 (2.03)	6.71 (2.01)	
	Machinery/electrical	5.50 (2.99)	3.00 (1.07)	4.06 (1.22)	4.93 (1.50)	
	Transport/other manuf.	9.25 (2.92)	0.68 (0.72)	3.35 (0.97)	5.25 (1.46)	

Notes: This table reports the OECD ICIO 1995–2007 baseline, imposing $\varepsilon_{ps}(\lambda) = \theta_{s0} + \theta_{g(s)1}\lambda$ in the 19 manufacturing sectors and using the ISIC two-digit tariff95 measure from the Global Tariff Database. Specification (A) estimates one manufacturing slope; (B) estimates consumer- and producer-manufacturing slopes; and (C) estimates seven manufacturing-aggregate slopes. Each sector has its own intercept. The column $\hat{\theta}_1$ reports the slope for the aggregate shown. For each aggregate, $\hat{\varepsilon}_p$ is the unweighted mean across sectors of the fitted sectoral elasticities evaluated at each sector's own p th percentile of the initial domestic expenditure share. Each specification uses 1,510 destination-sector observations. Standard errors are clustered by destination country and reported in parentheses.

Table: OECD ICIO Estimates of the Home-Foreign Trade Elasticity, 1995–2007

Estimation Results: GTAP Baseline

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	2.39 (0.82)	2.55 (0.51)	3.04 (0.57)	3.48 (0.66)	0.30
(B)	Consumer manuf.	2.12 (0.90)	3.01 (0.59)	3.47 (0.63)	3.88 (0.71)	0.30
	Producer manuf.	2.71 (0.93)	2.02 (0.48)	2.55 (0.55)	3.00 (0.65)	
(C)	Food, bev., tob.	2.04 (1.43)	3.95 (0.81)	4.40 (0.82)	4.79 (0.91)	0.30
	Textiles/apparel/leather	1.76 (0.96)	1.70 (0.40)	2.10 (0.42)	2.55 (0.56)	
	Wood/paper/printing	1.93 (1.30)	0.82 (0.34)	1.18 (0.38)	1.47 (0.50)	
	Petrol./chem./pharm./rubber	2.34 (1.05)	1.68 (0.41)	2.30 (0.51)	2.66 (0.62)	
	Minerals/metals	2.79 (1.34)	2.63 (0.80)	3.14 (0.81)	3.64 (0.89)	
	Machinery/electrical	3.21 (1.21)	2.30 (0.61)	2.69 (0.62)	3.17 (0.67)	
	Transport/other manuf.	3.64 (1.15)	1.92 (0.51)	2.61 (0.55)	3.26 (0.66)	

Notes: This table reports results using GTAP data and the GTAP-sector tariff95 measure from the Global Tariff Database over the single 2004–2014 time change, imposing $\varepsilon_{ns}(\lambda) = \theta_{s0} + \theta_{g(s)1}\lambda$. Specification (A) estimates one manufacturing slope; (B) estimates consumer- and producer-manufacturing slopes; and (C) estimates seven manufacturing-aggregate slopes. Each sector has its own intercept. The column $\hat{\theta}_1$ reports the slope for the aggregate shown. For each aggregate, $\hat{\varepsilon}_p$ is the unweighted mean across sectors of the fitted sectoral elasticities evaluated at each sector's own p th percentile of the initial domestic expenditure share. Each specification uses 3,240 destination–sector observations. Standard errors are clustered by destination country and reported in parentheses.

Table: GTAP Estimates of the Home-Foreign Trade Elasticity, 2004–2014

Estimator Validation and Empirical Sensitivity

Monte Carlo I: Estimator does not generate positive slopes in standard EK/Armington economies

Monte Carlo II: Estimator correctly identifies positive slopes when true schedules are linear

[Details](#)

1. Alternative samples + restrictions

● Non-manufacturing goods	OECD GTAP
● Country characteristics in intercepts	OECD GTAP
● Intercepts pooled by sector aggregate	OECD GTAP
● Sector-specific slopes	OECD GTAP

2. Tariff construction

● At least 50% tariff coverage	OECD GTAP
● Trade-weighted tariffs	OECD GTAP
● MFN tariffs	OECD GTAP

3. Functional form + influence

● Quadratic schedule	OECD GTAP
● Leave-one-country/sector-out	OECD GTAP

4. Positivity, finite changes + period

● Positivity restrictions	OECD GTAP
● Exact finite changes	OECD GTAP
● OECD ICIO, 2004–2013	OECD

Magnitudes typically change modestly, with some appreciable changes. The common slope remains positive in every sensitivity retaining a linear common schedule.

Endogeneity I: Correlated Measurement Error

- $\lambda_{nns,0}$ and the initial home-foreign expenditure ratio use the same expenditure data
- Measurement error can therefore be mechanically correlated across the dependent variable and the slope regressor
- Use 1995 data to instrument an estimating equation based on changes from 1996 to 2007

	$\hat{\theta}_1$	$\hat{\varepsilon}_{50}$
Without instruments	4.87	5.66
With lagged instruments	5.81	7.03

- **Similar magnitudes suggest correlated M.E. is not driving the estimates**

Endogeneity II: The Price Shifter

- Full estimation excluding $s \Rightarrow \hat{C}^{(-s)} \Rightarrow \hat{Q}_{ns}^{(-s)}$ for held-out sector (for each sector redo full baseline)
- Instrument $\hat{Q}_{ns}^{(-s)}$ w/ trade agreement-implied tariff changes using 2SLS (b/c cost change is fixed)
 - ▶ 2004 EU enlargement tariff changes only (all others set to zero)
 - ▶ + 2007 EU enlargement + EU-Türkiye + Gulf Cooperation Council + West African Economic/Monetary Union
- Instrument $\lambda_{nns,0} \hat{Q}_{ns}^{(-s)}$ w/ trade agreement-implied tariff changes interacted w/ $\lambda_{nns,0}$
- Destination-cluster bootstrap re-estimates $-s$ country costs + IV model in every draw

	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$
Baseline	5.45	4.01	5.12	5.88
2004 EU enlargement	8.83	3.62	5.42	6.65
	$[-1.77, 18.08]$			
Broader agreements	10.75	2.76	4.96	6.45
	$[1.20, 14.79]$			

Brackets report 95 percent destination-cluster bootstrap confidence intervals for the slope

- The fitted-elasticity profiles are broadly comparable in magnitude; both IV estimates imply a steeper increase across the distribution

Quantification

Production and Final Uses

- **Sectors:** manufacturing + non-manufacturing goods sectors; remaining sectors form a non-traded composite
- **Production:** Every sector in country n has value-added share β_n and uses the same aggregate composite as its intermediate input

$$c_{ns} = a_{ns}^{-1} w_n^{\beta_n} P_n^{1-\beta_n}$$

- **Final uses:** Household consumption, government purchases, and investment are current final demands; capital stocks are fixed
- **Intermediate inputs:** The scalar multiplier $1/\beta_n$ amplifies gains from final expenditure; median $\beta_n = 0.48$ in OECD ICIO and $= 0.42$ in GTAP

Details

Preferences

- **Sectoral cost of autarky:**

$$\hat{e}_{ns} = \exp \left(\int_{\lambda_{nns}^T}^1 \frac{d \ln \lambda}{\varepsilon_{ns}(\lambda)} \right)$$

- **Lower tier:** CES across sectors within manufacturing and non-manufacturing goods, with $\sigma_L = 2$
- **Upper tier:** CES across manufacturing, non-manufacturing goods, and non-goods, with $\sigma_U = 0.6$

$$\ln \frac{W_n^T}{W_n^A} = \frac{1}{\beta_n(1 - \sigma_U)} \ln \left[\alpha_{nM} \hat{E}_{nM}^{1-\sigma_U} + \alpha_{nG} \hat{E}_{nG}^{1-\sigma_U} + \alpha_{n0} \right]$$

Quantitative Results

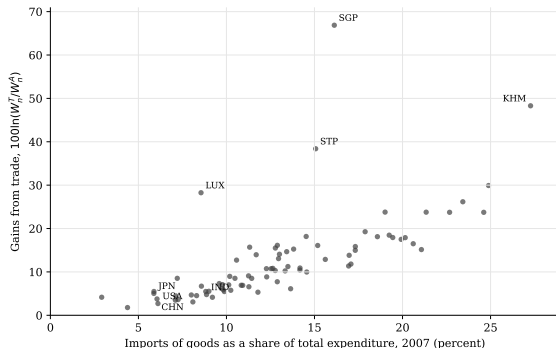
OECD ICIO gains from trade: $100 \ln(W_n^T / W_n^A)$

	p10	p25	Median	Mean	p75	p90
Specification (A)	4.45	6.01	10.34	12.53	15.64	23.15
Specification (B)	4.44	6.05	10.59	12.82	15.91	23.74
Specification (C)	4.34	6.14	10.56	12.80	15.92	23.47

- **Specification (B):** median gains are 10.59 log points and mean gains are 12.82
- Gains are substantial and heterogeneous across countries
- Results are very similar across the three slope-pooling specifications

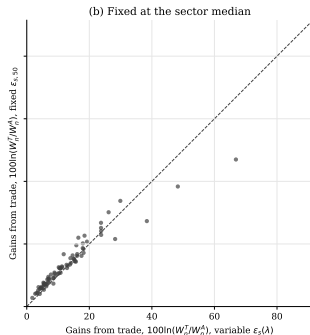
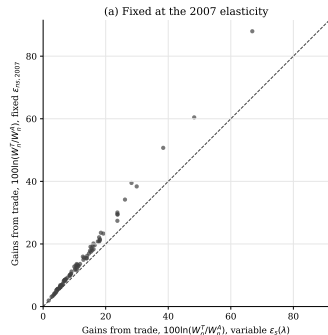
GTAP

Gains from Trade and Initial Import Shares



- **Trade exposure:** Correlation is 0.67; countries w/ $\uparrow$ initial import shares generally gain more
- **Composition:** Countries with similar aggregate import shares can differ because sectoral shares, elasticity schedules, and value-added shares differ
 - ▶ Singapore has $\lambda_{nns} = 0$ (and, therefore low elasticity) in several non-manufacturing goods s
 - ▶ Cambodia has high expenditure share in sectors with low λ_{nns} (and therefore low elasticity)

Variable and Fixed home-foreign Elasticities



- **Initial value:** Holding each country-sector elasticity fixed at its 2007 value raises gains for every country
- **Mean rises from 12.82 to 15.60 log points**
- **Sector median:** The ordering varies with countries' initial sectoral domestic shares

Within-sector mechanism: A lower initial domestic share means greater trade exposure, a lower starting elasticity, and a longer portion of the schedule traversed on the path to autarky. GTAP

Sensitivity

σ_L	σ_U	p25	Median	Mean	p75
2	0.6	6.05	10.59	12.82	15.91
1	0.6	7.57	18.65	∞	∞
3	0.6	5.58	9.37	10.68	13.70
2	1	6.00	10.26	11.99	15.33

- Raising σ_L allows substitution away from sectors with the largest price-index increases and lowers measured gains
- With $\sigma_L = 1$, zero domestic expenditure makes gains infinite for 20 countries
- Raising σ_U from 0.6 to one has a more modest effect on the median

Conclusion

Conclusion

- **Negative:** The elasticity estimated from substitution across foreign sources is either irrelevant for GFT or only relevant in extremely restricted models
- **Positive:** The welfare-relevant home-foreign elasticity schedule is low dimensional and can be estimated directly
- **Evidence:** In both data sets, fitted elasticities generally rise with domestic expenditure shares, including under extensive sensitivity
- **Quantification:** In OECD ICIO, mean GFT are 12.82 log points, compared with 15.60 when each elasticity is fixed at its initial value

Appendix

$$\Delta \ln \frac{X_{nns}}{X_{nFs}} = \varepsilon_s(\lambda_{nns,0}) Q_{ns}(\mathbf{C}) + u_{ns}, \quad Q_{ns}(\mathbf{C}) = Q_{ns}^{\mathcal{T}} + Q_{ns}^{\mathcal{C}}(\mathbf{C})$$

For each $\lambda \in \text{supp}(\lambda_{nns,0} \mid s)$:

Condition on $\lambda_{nns,0} = \lambda$, $Q_{ns}^{\mathcal{C}}(\mathbf{C})$, predetermined shares and controls, and sector s . **Assume**

$$\underbrace{\text{Var}(Q_{ns}^{\mathcal{T}} \mid \cdot) > 0}_{\substack{\text{Relevance} \\ \text{(rank condition)}}} \quad \text{and} \quad \underbrace{\text{Cov}(Q_{ns}^{\mathcal{T}}, u_{ns} \mid \cdot) = 0}_{\text{Exclusion}}$$

These conditions identify

$$\varepsilon_s(\lambda) = \frac{\text{Cov}\left(\Delta \ln \frac{X_{nns}}{X_{nFs}}, Q_{ns}^{\mathcal{T}} \mid \cdot\right)}{\text{Var}(Q_{ns}^{\mathcal{T}} \mid \cdot)} \quad \text{for every } \lambda \in \text{supp}(\lambda_{nns,0} \mid s).$$

- **Beyond the support of domestic expenditure shares in sector s , GFT relies on functional form to extrapolate the elasticity schedule**

Appendix: Moment Conditions

Nonlinear regression function: $m_{ns}(\gamma) = [\theta_{s0} + \theta_{g(s)1} \lambda_{nns,0}] \left[Q_{ns}^T + \sum_{i \neq n} \mu_{nis,0} (C_i - C_n) \right]$

Let $Y_{ns} \equiv \Delta \log(X_{nns}/X_{nFs})$, $u_{ns} = Y_{ns} - m_{ns}(\gamma_0)$, $D_{ns}(\gamma_0) \equiv \partial m_{ns}(\gamma_0)/\partial \gamma$

NLLS imposes $\mathbb{E}[D_{ns}(\gamma_0)u_{ns}] = 0$

Writing $\varepsilon_{ns,0} \equiv \theta_{s0} + \theta_{g(s)1} \lambda_{nns,0}$, these moments are

$$\begin{aligned} \frac{\partial}{\partial \theta_{\ell 0}} : \quad & \mathbb{E}[\mathbf{1}\{s = \ell\} Q_{ns}(\mathbf{C}) u_{ns}] = 0 \quad \text{for every sector } \ell \\ \frac{\partial}{\partial \theta_{g1}} : \quad & \mathbb{E}[\mathbf{1}\{g(s) = g\} \lambda_{nns,0} Q_{ns}(\mathbf{C}) u_{ns}] = 0 \quad \text{for every } g \\ \frac{\partial}{\partial C_j} : \quad & \mathbb{E} \left[\varepsilon_{ns,0} \left(\mathbf{1}\{n \neq j\} \mu_{njs,0} - \mathbf{1}\{n = j\} \sum_{i \neq n} \mu_{nis,0} \right) u_{ns} \right] = 0 \quad \text{for every } j \neq \text{USA} \end{aligned}$$

- For each candidate elasticity vector θ , solve by least squares for $\hat{C}(\theta)$ with $C_{\text{USA}} = 0$
- Substitute $\hat{C}(\theta)$ into sum of squared residuals $\rightarrow$ minimize profiled NLLS objective over θ
- Use the Broyden–Fletcher–Goldfarb–Shanno (BFGS) quasi-Newton algorithm
 - ▶ At the current θ , calculate the gradient of the profiled objective
 - ▶ BFGS uses changes btw successive iterations in θ and gradient to approximate inverse Hessian
 - ▶ It combines this approximation with the current gradient to choose a descent direction and then determines how far to move along that direction
- Evaluate the analytic gradient holding $\hat{C}(\theta)$ fixed: the envelope term vanishes because the conditional least-squares first-order conditions hold
- BFGS does not require Jacobian of profiled residual vector

For any finite $m \geq m_0 \geq 1$:

integrate definition of ε_n

$$\ln \frac{\lambda_{nn}(m)}{1 - \lambda_{nn}(m)} = \ln \frac{\lambda_{nn}(m_0)}{1 - \lambda_{nn}(m_0)} + \int_{m_0}^m \varepsilon_n(z) d \ln z.$$

Initial domestic expenditure share, $\lambda_{nn}(m_0)$, and elasticity schedule, $\varepsilon_n(m)$, determine path of domestic expenditure shares

For finite gains, assume home good alone can deliver one unit of utility. Then

$$\ln \frac{W_n^T}{W_n^A} = \int_1^\infty [1 - \lambda_{nn}(m)] d \ln m.$$

Calculating gains from trade requires only the initial domestic expenditure share and the home-foreign elasticity schedule along the proportional ray

Foreign relative-price growth along a path Γ from trade ($t = 1$) to autarky:

$$\alpha_{ni}(t) \equiv \frac{d \ln \pi_{ni}(t)}{dt} \quad (i \neq n) \quad \text{Baseline: } \alpha_{ni}(t) = \frac{d \ln m(t)}{dt}$$

With $\mu_{ni}(t) \equiv X_{ni}(t)/X_{nF}(t)$, the expenditure equation becomes

$$\frac{d \ln H_n(t)}{dt} = \sum_{i \neq n} \lambda_{ni}(t) \alpha_{ni}(t) = [1 - \lambda_{nn}(t)] \sum_{i \neq n} \mu_{ni}(t) \alpha_{ni}(t).$$

$$\text{Define } z'_n(t) \equiv \sum_{i \neq n} \mu_{ni}(t) \alpha_{ni}(t) \quad \text{Baseline: } z'_n(t) = \frac{d \ln m(t)}{dt}$$

The home-foreign elasticity along Γ gives

$$\varepsilon_n^\Gamma(t) \equiv \frac{d \ln [X_{nn}(t)/X_{nF}(t)]/dt}{z'_n(t)} \quad \Rightarrow \quad z'_n(t) = \frac{1}{[1 - \lambda_{nn}(t)] \varepsilon_n^\Gamma(t)} \frac{d \ln \lambda_{nn}(t)}{dt}.$$

Assume $z'_n(t) > 0$ and $\varepsilon_n^\Gamma(t) > 0$, so $\lambda_{nn}(t)$ can index the path. Combining gives

$$\frac{d \ln H_n(t)}{dt} = \frac{1}{\varepsilon_n^\Gamma(t)} \frac{d \ln \lambda_{nn}(t)}{dt} \quad \Rightarrow \quad \ln \frac{W_n^T}{W_n^A} = \int_{\lambda_{nn}^T}^1 \frac{d \ln \lambda}{\varepsilon_n^\Gamma(\lambda)}.$$

- **Construct policy-free auxiliary economy**
 - ▶ Fix primitives and the preferences of every $k \neq n$ but give n Leontief preferences
 - ▶ Fix n 's consumption proportions to match Hicksian proportions w/ original preferences at target prices
- **Auxiliary equilibrium (before policy wedges are chosen) $\Rightarrow$**
 - ▶ feasible allocation, supporting unit costs, balanced trade
- **Use these costs to define (additive) wedges in original economy that implement the target prices; verify that the auxiliary allocation remains an equilibrium under those policies**

$$t_{ni}(m) = c_n(m)\pi_{ni}(m) - \mathcal{D}_{ni}c_i(m)$$

Auxiliary eqm supplies feasible allocation, supporting unit costs, and balanced trade before wedges are defined; this is the role played by planner stage in an implementation argument

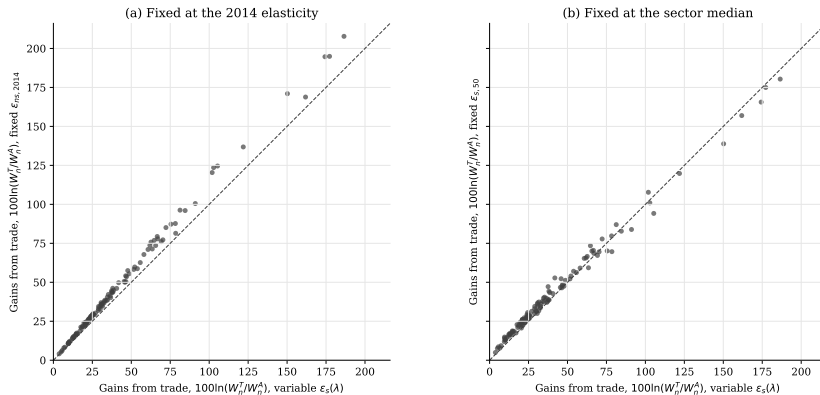
Leontief preferences fix the target proportions independently of costs

- Every production sector in country n has the same value-added share β_n
- Every sector uses the final-consumption aggregate as its intermediate-input composite
- A one-percent rise in the final-expenditure price index raises production costs by $1 - \beta_n$ percent
- This feedback amplifies the final-expenditure gains by $1/\beta_n$

$$\ln \frac{W_n^T}{W_n^A} = \frac{1}{\beta_n} \times \text{change in the final-expenditure price index}$$

	p10	p25	Median	Mean	p75	p90
Specification (A)	12.20	20.44	30.77	41.65	51.79	78.75
Specification (B)	12.16	20.42	30.61	41.61	51.74	78.77
Specification (C)	11.42	20.08	30.63	42.09	52.20	80.55

- Entries are $100 \ln(W_n^T / W_n^A)$ across 120 countries
- The three specifications vary the pooling restriction on schedule slopes



- Median GFT are 30.61 log points under variable schedules and 34.79 when each country-sector elasticity is held fixed at its initial value

- Simulate tariff shocks in new many-industry extended Armington model
 - ▶ nested homothetic CDE-CES expenditure system
 - ▶ separate home-foreign vs. across foreign substitution
 - ▶ home-foreign elasticity schedule is exactly linear
- Calibrate four (restricted elasticity schedules) versions of the model to same initial eqm
 - ▶ Slopes=0; intercepts= across-foreign elasticity (standard many-sector Armington/EK) [Table](#)
 - ▶ Single slope across all sectors [Table](#)
 - ▶ Single slope within each of the 7 groups of Specification C [Table](#)
 - ▶ One slope per individual sector [Table](#)
- For each, choose a random sample of n triplets ($w/ i \neq n$) to shock tariffs randomly
- Apply baseline estimator
- Repeat 200 times and present average estimates and distribution

Spec.	Aggregate	True sector slope(s)	Mean $\hat{\theta}_1$	5th pct.	95th pct.
(A)	All manufacturing	0.00	0.001	-0.004	0.006
(B)	Consumer manuf.	0.00	0.002	-0.004	0.007
	Producer manuf.	0.00	0.001	-0.008	0.009
(C)	Food, bev., tob.	0.00	0.003	-0.003	0.008
	Textiles/apparel/leather	0.00	-0.001	-0.022	0.016
	Wood/paper/printing	0.00	0.002	-0.007	0.011
	Petrol./chem./pharm./rubber	0.00	-0.001	-0.018	0.015
	Minerals/metals	0.00	0.000	-0.012	0.013
	Machinery/electrical	0.00	0.001	-0.010	0.011
	Transport/other manuf.	0.00	0.002	-0.007	0.011

Notes: Every sector has $\theta_{s1} = 0$ and $\rho_s = \theta_{s0}$. The table reports results from Specifications (A), (B), and (C) across 200 independently drawn tariff shocks, holding the realized 40-country, 27-sector Armington economy fixed. The mean, 5th percentile, and 95th percentile refer to the distribution of the estimated pooled slope across tariff shocks. The true-slope column reports the common true slope among sectors in the row when one exists and their range otherwise. Every regression uses the same unrestricted NLLS estimation routine as the real-data tables, including the jointly estimated country cost shifters.

Spec.	Aggregate	True sector slope(s)	Mean $\hat{\theta}_1$	5th pct.	95th pct.
(A)	All manufacturing	4.00	4.040	4.030	4.050
(B)	Consumer manuf.	4.00	4.035	4.024	4.045
	Producer manuf.	4.00	4.046	4.030	4.063
(C)	Food, bev., tob.	4.00	4.033	4.021	4.045
	Textiles/apparel/leather	4.00	4.043	4.006	4.079
	Wood/paper/printing	4.00	4.030	4.013	4.051
	Petrol./chem./pharm./rubber	4.00	4.049	4.015	4.080
	Minerals/metals	4.00	4.061	4.034	4.089
	Machinery/electrical	4.00	4.026	4.003	4.048
	Transport/other manuf.	4.00	4.045	4.028	4.064

Notes: Every sector has $\theta_{s1} = 4$, so Specification (A) matches the data-generating pooling structure. The table reports results from Specifications (A), (B), and (C) across 200 independently drawn tariff shocks, holding the realized 40-country, 27-sector Armington economy fixed. The mean, 5th percentile, and 95th percentile refer to the distribution of the estimated pooled slope across tariff shocks. The true-slope column reports the common true slope among sectors in the row when one exists and their range otherwise. Every regression uses the same unrestricted NLLS estimation routine as the real-data tables, including the jointly estimated country cost shifters.

Spec.	Aggregate	True sector slope(s)	Mean $\hat{\theta}_1$	5th pct.	95th pct.
(A)	All manufacturing	[2.50, 5.50]	3.745	3.673	3.815
(B)	Consumer manuf.	[2.50, 5.50]	3.256	3.173	3.360
	Producer manuf.	[3.50, 5.50]	4.380	4.322	4.438
(C)	Food, bev., tob.	2.50	2.523	2.514	2.533
	Textiles/apparel/leather	3.00	3.031	2.998	3.064
	Wood/paper/printing	3.50	3.528	3.512	3.546
	Petrol./chem./pharm./rubber	4.00	4.048	4.017	4.080
	Minerals/metals	4.50	4.568	4.540	4.597
	Machinery/electrical	5.00	5.029	5.003	5.054
	Transport/other manuf.	5.50	5.559	5.538	5.581

Notes: Slopes equal 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, and 5.5 across the seven manufacturing aggregates, so Specification (C) matches the data-generating pooling structure. The table reports results from Specifications (A), (B), and (C) across 200 independently drawn tariff shocks, holding the realized 40-country, 27-sector Armington economy fixed. The mean, 5th percentile, and 95th percentile refer to the distribution of the estimated pooled slope across tariff shocks. The true-slope column reports the common true slope among sectors in the row when one exists and their range otherwise. Every regression uses the same unrestricted NLLS estimation routine as the real-data tables, including the jointly estimated country cost shifters.

Appendix: Monte Carlo, Sector-Specific Slopes

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Spec.	Aggregate	True sector slope(s)	Mean $\hat{\theta}_1$	5th pct.	95th pct.
(A)	All manufacturing	[2.29, 5.95]	4.139	4.063	4.215
(B)	Consumer manuf.	[2.46, 5.93]	4.107	4.009	4.210
	Producer manuf.	[2.29, 5.95]	4.178	4.068	4.313
(C)	Food, bev., tob.	[2.65, 5.93]	4.702	4.624	4.778
	Textiles/apparel/leather	[3.38, 4.38]	3.531	3.481	3.573
	Wood/paper/printing	[3.26, 5.95]	4.828	4.601	5.109
	Petrol./chem./pharm./rubber	[2.29, 5.76]	4.071	3.787	4.384
	Minerals/metals	[2.58, 4.88]	3.471	3.342	3.604
	Machinery/electrical	[3.49, 5.46]	4.208	4.120	4.292
	Transport/other manuf.	[2.46, 4.61]	3.373	3.240	3.505

Notes: Every sector has a distinct positive slope. The table reports results from Specifications (A), (B), and (C) across 200 independently drawn tariff shocks, holding the realized 40-country, 27-sector Armington economy fixed. The mean, 5th percentile, and 95th percentile refer to the distribution of the estimated pooled slope across tariff shocks. The true-slope column reports the common true slope among sectors in the row when one exists and their range otherwise. Every regression uses the same unrestricted NLLS estimation routine as the real-data tables, including the jointly estimated country cost shifters.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All goods	4.98 (1.55)	3.85 (0.87)	4.93 (1.04)	5.58 (1.17)	0.18
(B)	Consumer manuf.	6.92 (2.00)	2.59 (0.65)	4.05 (0.86)	4.99 (1.06)	0.19
	Producer manuf.	0.54 (2.50)	4.67 (1.25)	4.77 (1.10)	4.85 (1.13)	
	Non-manufacturing	7.69 (2.92)	4.08 (1.08)	5.99 (1.39)	6.83 (1.61)	
(C)	Food, bev., tob.	1.25 (3.31)	3.27 (0.73)	3.39 (0.70)	3.45 (0.74)	0.19
	Textiles/apparel/leather	10.79 (3.88)	0.94 (1.00)	3.87 (0.91)	5.55 (1.31)	
	Wood/paper/printing	0.64 (4.00)	3.01 (0.94)	3.11 (0.78)	3.17 (0.90)	
	Petrol./chem./pharm./rubber	3.81 (2.41)	4.39 (1.07)	5.07 (1.12)	5.50 (1.24)	
	Minerals/metals	-4.29 (3.95)	7.22 (1.93)	6.41 (1.54)	5.87 (1.44)	
	Machinery/electrical	4.95 (2.66)	2.47 (0.93)	3.41 (0.95)	4.20 (1.15)	
	Transport/other manuf.	9.10 (2.54)	0.08 (0.70)	2.71 (0.69)	4.58 (1.06)	
	Non-manufacturing	6.93 (2.75)	4.30 (1.10)	6.02 (1.35)	6.77 (1.55)	

Notes: This table replicates Table 1, but adds the seven non-manufacturing goods sectors with matched tariff data. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All goods	3.73 (1.56)	-0.56 (0.31)	0.32 (0.25)	1.09 (0.51)	0.20
(B)	Consumer manuf.	3.30 (1.43)	0.24 (0.27)	0.94 (0.43)	1.59 (0.68)	0.20
	Producer manuf.	4.04 (1.88)	-0.03 (0.24)	0.75 (0.40)	1.42 (0.68)	
	Non-manufacturing	3.94 (1.72)	-1.64 (0.76)	-0.53 (0.40)	0.45 (0.43)	
(C)	Food, bev., tob.	4.04 (1.75)	0.20 (0.33)	1.09 (0.51)	1.85 (0.80)	0.20
	Textiles/apparel/leather	2.57 (1.31)	0.17 (0.31)	0.76 (0.39)	1.41 (0.65)	
	Wood/paper/printing	3.95 (1.84)	-1.27 (0.51)	-0.54 (0.31)	0.06 (0.36)	
	Petrol./chem./pharm./rubber	1.97 (1.37)	0.49 (0.32)	1.00 (0.47)	1.30 (0.64)	
	Minerals/metals	4.92 (2.35)	0.16 (0.37)	1.05 (0.56)	1.93 (0.93)	
	Machinery/electrical	4.11 (2.05)	0.09 (0.32)	0.60 (0.36)	1.22 (0.57)	
	Transport/other manuf.	4.11 (1.85)	0.12 (0.33)	0.90 (0.46)	1.64 (0.73)	
	Non-manufacturing	3.89 (1.69)	-1.61 (0.74)	-0.51 (0.39)	0.45 (0.42)	

Notes: This table replicates Table 2, but adds the 17 non-manufacturing goods sectors. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	5.92 (1.85)	3.26 (0.80)	4.47 (1.02)	5.29 (1.22)	0.28
(B)	Consumer manuf.	8.71 (2.33)	2.37 (0.60)	4.19 (0.90)	5.38 (1.16)	0.28
	Producer manuf.	3.59 (2.46)	4.39 (1.16)	5.10 (1.31)	5.61 (1.50)	
(C)	Food, bev., tob.	16.60 (6.14)	0.86 (0.50)	2.52 (0.70)	3.23 (0.91)	0.29
	Textiles/apparel/leather	16.15 (5.09)	-1.13 (0.96)	3.25 (0.80)	5.76 (1.48)	
	Wood/paper/printing	0.79 (2.39)	2.55 (0.81)	2.67 (0.76)	2.74 (0.81)	
	Petrol./chem./pharm./rubber	8.40 (3.11)	2.73 (0.85)	4.21 (1.19)	5.16 (1.48)	
	Minerals/metals	0.56 (2.46)	4.16 (1.34)	4.26 (1.37)	4.33 (1.47)	
	Machinery/electrical	7.53 (2.58)	0.77 (0.59)	2.22 (0.79)	3.42 (1.10)	
	Transport/other manuf.	10.32 (3.63)	-1.02 (0.88)	1.96 (0.64)	4.08 (1.22)	

Notes: This table replicates Table 1, but allows sector elasticity intercepts to vary with 1995 PPP GDP per capita, population, remoteness, and landlocked status. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	3.62 (1.06)	1.70 (0.39)	2.44 (0.48)	3.10 (0.62)	0.31
(B)	Consumer manuf.	3.45 (1.11)	2.28 (0.48)	3.01 (0.58)	3.69 (0.72)	0.31
	Producer manuf.	3.87 (1.24)	1.07 (0.38)	1.83 (0.43)	2.47 (0.56)	
(C)	Food, bev., tob.	3.86 (1.41)	3.19 (0.68)	4.04 (0.78)	4.77 (0.95)	0.31
	Textiles/apparel/leather	2.33 (1.11)	1.19 (0.45)	1.72 (0.45)	2.31 (0.59)	
	Wood/paper/printing	2.17 (1.63)	0.15 (0.45)	0.54 (0.38)	0.88 (0.48)	
	Petrol./chem./pharm./rubber	3.14 (1.15)	1.03 (0.33)	1.85 (0.42)	2.33 (0.55)	
	Minerals/metals	3.77 (1.42)	1.61 (0.59)	2.29 (0.62)	2.97 (0.74)	
	Machinery/electrical	4.89 (1.58)	1.09 (0.50)	1.70 (0.47)	2.43 (0.53)	
	Transport/other manuf.	5.15 (1.53)	0.95 (0.48)	1.92 (0.46)	2.84 (0.59)	

Notes: This table replicates Table 2, but allows sector elasticity intercepts to vary with 2004 PPP GDP per capita, population, remoteness, and landlocked status. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	1.57 (1.34)	3.20 (1.00)	3.64 (1.13)	3.91 (1.26)	0.17
(B)	Consumer manuf.	3.04 (1.26)	2.37 (0.76)	3.25 (0.97)	3.86 (1.17)	0.18
	Producer manuf.	-1.40 (2.03)	3.90 (1.31)	3.54 (1.24)	3.33 (1.30)	
(C)	Food, bev., tob.	3.90 (3.21)	3.07 (0.86)	3.46 (0.88)	3.62 (0.92)	0.21
	Textiles/apparel/leather	11.75 (4.35)	0.75 (0.91)	3.94 (1.10)	5.76 (1.64)	
	Wood/paper/printing	-1.68 (3.55)	3.14 (1.12)	2.90 (0.95)	2.74 (0.97)	
	Petrol./chem./pharm./rubber	2.09 (2.16)	4.12 (1.42)	4.43 (1.50)	4.80 (1.67)	
	Minerals/metals	-5.89 (4.73)	6.44 (2.32)	5.07 (1.95)	4.34 (1.99)	
	Machinery/electrical	3.42 (2.11)	3.15 (1.25)	3.84 (1.36)	4.54 (1.59)	
	Transport/other manuf.	4.76 (2.20)	0.95 (0.76)	2.61 (0.85)	3.74 (1.23)	

Notes: This table replicates Table 1, but pools intercepts within manufacturing aggregates. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	2.45 (0.70)	2.55 (0.53)	3.23 (0.64)	3.87 (0.77)	0.26
(B)	Consumer manuf.	2.59 (0.78)	2.67 (0.54)	3.56 (0.70)	4.20 (0.85)	0.26
	Producer manuf.	2.09 (0.96)	2.17 (0.54)	2.66 (0.61)	3.14 (0.74)	
(C)	Food, bev., tob.	1.51 (0.89)	3.91 (0.76)	4.38 (0.86)	4.62 (0.94)	0.27
	Textiles/apparel/leather	2.60 (1.13)	1.73 (0.44)	2.24 (0.48)	3.03 (0.69)	
	Wood/paper/printing	1.22 (1.44)	1.23 (0.32)	1.46 (0.39)	1.64 (0.55)	
	Petrol./chem./pharm./rubber	2.07 (1.11)	1.75 (0.42)	2.18 (0.49)	2.68 (0.67)	
	Minerals/metals	2.34 (1.59)	2.54 (0.87)	3.13 (0.80)	3.60 (0.89)	
	Machinery/electrical	4.66 (1.49)	2.26 (0.57)	2.81 (0.65)	3.72 (0.84)	
	Transport/other manuf.	2.96 (0.93)	1.94 (0.52)	2.61 (0.57)	3.28 (0.68)	

Notes: This table replicates Table 2, but pools intercepts within manufacturing aggregates. Standard errors are clustered by destination country and reported in parentheses.

Industry	$\hat{\theta}_{40}$	$\hat{\theta}_{41}$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$
C10T12 Food, beverages, tobacco	-2.53 (3.18)	7.11 (3.89)	3.00 (0.70)	3.71 (0.77)	4.01 (0.85)
C13T15 Textiles, apparel, leather	-4.06 (2.27)	12.21 (4.06)	0.88 (0.92)	4.20 (0.98)	6.09 (1.46)
C16 Wood products	5.58 (3.78)	-2.33 (4.70)	4.14 (1.35)	3.65 (1.20)	3.46 (1.35)
C17_18 Paper and printing	3.17 (3.17)	-0.98 (4.12)	2.53 (0.78)	2.42 (0.65)	2.32 (0.78)
C19 Coke and refined petroleum	2.81 (3.19)	5.05 (4.46)	5.74 (1.92)	6.93 (2.20)	7.41 (2.44)
C20 Chemicals	2.34 (2.28)	7.21 (4.12)	5.34 (1.84)	6.62 (2.09)	7.36 (2.33)
C21 Pharmaceuticals	2.74 (3.68)	2.19 (5.04)	3.88 (1.43)	4.27 (1.14)	4.53 (1.29)
C22 Rubber and plastics	3.26 (2.91)	-0.74 (4.35)	2.88 (1.04)	2.79 (0.91)	2.69 (1.09)
C23 Other non-metallic minerals	3.41 (2.72)	-1.61 (3.14)	2.16 (0.70)	2.03 (0.67)	1.95 (0.70)
C24A Iron and steel	7.44 (3.13)	1.83 (3.38)	8.21 (2.45)	8.61 (2.37)	8.90 (2.46)
C24B Non-ferrous metals	14.27 (6.43)	0.40 (7.46)	14.39 (5.06)	14.51 (4.53)	14.59 (4.73)
C25 Fabricated metals	6.00 (2.66)	-4.69 (3.21)	3.35 (1.07)	2.60 (0.81)	2.09 (0.79)
C26 Computer, electronic, optical	0.63 (1.52)	5.53 (4.56)	1.47 (1.12)	2.34 (1.07)	3.31 (1.50)
C27 Electrical equipment	2.14 (1.92)	6.16 (3.90)	4.05 (1.43)	5.50 (1.66)	6.23 (1.94)
C28 Machinery and equipment	2.74 (1.77)	4.88 (4.59)	3.40 (1.48)	4.29 (1.46)	5.18 (1.87)
C29 Motor vehicles	-1.20 (1.32)	9.45 (3.30)	0.08 (0.97)	2.75 (0.78)	5.32 (1.38)
C301 Ships and boats	-1.99 (2.96)	12.35 (5.60)	-0.25 (2.42)	4.84 (2.09)	8.17 (3.03)
C302T309 Rail, aircraft, other transport	-0.25 (1.26)	8.90 (3.62)	0.96 (1.08)	3.90 (1.46)	5.71 (2.04)
C31T33 Furniture, other manufacturing, repair	-0.99 (1.78)	4.26 (2.34)	1.80 (0.74)	2.34 (0.77)	2.67 (0.84)

Notes: This table replicates Table 1, but allows every manufacturing sector to have its own intercept and slope. Standard errors are clustered by destination country and reported in parentheses. The specification uses 1,510 observations and has $R^2 = 0.28$.

Industry	$\hat{\theta}_{s0}$	$\hat{\theta}_{s1}$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$
19 Cattle meat	-8.38 (4.65)	15.37 (6.15)	3.10 (1.12)	5.29 (1.47)	6.66 (1.87)
20 Other meat	2.62 (1.25)	3.02 (2.16)	4.63 (1.07)	5.30 (1.39)	5.55 (1.52)
21 Vegetable oils	3.52 (1.10)	1.34 (1.38)	3.85 (0.97)	4.16 (0.95)	4.48 (1.04)
22 Dairy products	2.10 (1.08)	1.48 (1.40)	3.06 (0.67)	3.36 (0.75)	3.50 (0.82)
23 Processed rice	2.52 (1.39)	3.93 (2.51)	3.00 (1.22)	4.02 (1.07)	6.15 (1.79)
24 Sugar/molasses	4.49 (1.76)	-0.90 (3.15)	4.28 (1.32)	3.92 (1.38)	3.67 (1.95)
25 Other food	2.94 (1.65)	0.64 (2.00)	3.35 (0.77)	3.43 (0.74)	3.50 (0.79)
26 Beverages/tobacco	1.26 (1.68)	1.53 (2.07)	2.27 (0.64)	2.54 (0.62)	2.67 (0.69)
27 Textiles	-0.13 (0.50)	1.39 (1.34)	0.14 (0.34)	0.36 (0.31)	0.62 (0.44)
28 Wearing apparel	1.04 (0.59)	3.68 (1.27)	1.92 (0.51)	3.04 (0.64)	4.09 (0.91)
29 Leather products	2.72 (0.73)	-0.04 (1.10)	2.71 (0.64)	2.71 (0.58)	2.69 (0.64)
30 Wood products	0.10 (0.94)	-0.10 (1.31)	0.05 (0.42)	0.03 (0.35)	0.01 (0.43)
31 Paper/printing	0.16 (0.87)	3.49 (1.65)	1.54 (0.51)	2.16 (0.56)	2.63 (0.69)
32 Petroleum/coke	1.32 (0.99)	-0.00 (1.84)	1.32 (0.86)	1.32 (0.98)	1.32 (1.27)
33 Chemicals	1.10 (0.63)	3.45 (1.36)	1.85 (0.57)	2.47 (0.62)	2.87 (0.70)
34 Pharmaceuticals	0.87 (0.46)	2.79 (1.20)	1.22 (0.40)	1.69 (0.40)	2.06 (0.47)
35 Rubber/plastics	1.24 (0.69)	1.96 (1.31)	1.93 (0.51)	2.29 (0.56)	2.57 (0.66)
36 Non-metallic minerals	0.11 (1.35)	2.96 (1.92)	1.85 (0.59)	2.25 (0.61)	2.56 (0.69)
37 Iron/steel	2.22 (1.12)	0.99 (1.79)	2.37 (0.93)	2.61 (0.75)	2.82 (0.77)
38 Non-ferrous metals	3.37 (2.62)	1.99 (3.82)	3.81 (1.96)	4.14 (1.60)	4.60 (1.44)
39 Fabricated metals	-0.06 (0.56)	4.61 (1.27)	1.95 (0.52)	2.77 (0.66)	3.57 (0.83)
40 Computers/electronics/optical	1.00 (0.60)	3.08 (1.52)	1.48 (0.52)	1.74 (0.52)	2.17 (0.58)
41 Electrical equipment	3.71 (1.38)	4.05 (2.04)	4.47 (1.27)	5.19 (1.27)	5.96 (1.38)
42 Machinery/equipment n.e.c.	0.27 (0.74)	2.13 (2.52)	0.44 (0.61)	0.67 (0.49)	0.93 (0.53)
43 Motor vehicles	1.40 (0.64)	3.04 (1.22)	1.65 (0.61)	2.21 (0.60)	2.82 (0.68)
44 Other transport equip.	0.37 (0.96)	5.34 (2.63)	1.29 (0.71)	2.31 (0.73)	3.31 (1.02)
45 Other manufacturing	0.73 (0.60)	3.26 (0.97)	2.15 (0.53)	2.78 (0.60)	3.27 (0.69)

Notes: This table replicates Table 2, but allows every manufacturing sector to have its own intercept and slope. Standard errors are clustered by destination country and reported in parentheses. The specification uses 3,240 observations and has $R^2 = 0.31$.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	5.46 (1.87)	4.01 (1.12)	5.12 (1.32)	5.88 (1.49)	0.26
(B)	Consumer manuf.	8.02 (2.43)	2.99 (0.82)	4.66 (1.13)	5.74 (1.39)	0.27
	Producer manuf.	2.11 (2.48)	5.15 (1.56)	5.57 (1.56)	5.87 (1.66)	
(C)	Food, bev., tob.	6.97 (3.93)	3.03 (0.74)	3.73 (0.80)	4.03 (0.88)	0.28
	Textiles/apparel/leather	12.31 (4.08)	0.84 (0.91)	4.18 (1.00)	6.10 (1.48)	
	Wood/paper/printing	-1.20 (4.14)	3.33 (1.03)	3.14 (0.87)	3.03 (0.98)	
	Petrol./chem./pharm./rubber	3.60 (2.79)	4.57 (1.31)	5.19 (1.45)	5.59 (1.61)	
	Minerals/metals	-1.45 (3.15)	7.16 (2.18)	6.89 (2.03)	6.70 (2.01)	
	Machinery/electrical	5.54 (2.99)	2.99 (1.07)	4.05 (1.22)	4.93 (1.51)	
	Transport/other manuf.	9.25 (2.92)	0.69 (0.71)	3.35 (0.97)	5.25 (1.46)	

Notes: This table replicates Table 1, but requires origins with observed tariff changes to account for at least 50 percent of initial foreign expenditure. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	2.45 (0.83)	2.56 (0.51)	3.06 (0.57)	3.51 (0.67)	0.30
(B)	Consumer manuf.	2.18 (0.91)	3.02 (0.59)	3.49 (0.63)	3.91 (0.72)	0.30
	Producer manuf.	2.80 (0.93)	2.02 (0.48)	2.57 (0.55)	3.03 (0.65)	
(C)	Food, bev., tob.	2.17 (1.44)	3.96 (0.81)	4.44 (0.82)	4.85 (0.92)	0.30
	Textiles/apparel/leather	1.74 (0.97)	1.74 (0.40)	2.13 (0.43)	2.57 (0.56)	
	Wood/paper/printing	1.88 (1.29)	0.84 (0.34)	1.19 (0.38)	1.48 (0.50)	
	Petrol./chem./pharm./rubber	2.55 (1.05)	1.70 (0.42)	2.37 (0.52)	2.76 (0.63)	
	Minerals/metals	2.89 (1.35)	2.62 (0.80)	3.14 (0.81)	3.66 (0.89)	
	Machinery/electrical	3.27 (1.22)	2.29 (0.61)	2.69 (0.62)	3.18 (0.67)	
	Transport/other manuf.	3.70 (1.15)	1.90 (0.51)	2.61 (0.55)	3.27 (0.66)	

Notes: This table replicates Table 2, but requires origins with observed tariff changes to account for at least 50 percent of initial foreign expenditure. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	4.52 (1.53)	3.43 (0.86)	4.35 (1.02)	4.99 (1.17)	0.27
(B)	Consumer manuf.	6.73 (1.99)	2.48 (0.63)	3.87 (0.86)	4.81 (1.08)	0.27
	Producer manuf.	1.58 (2.05)	4.38 (1.19)	4.69 (1.22)	4.91 (1.33)	
(C)	Food, bev., tob.	10.03 (4.75)	2.26 (0.62)	3.27 (0.71)	3.67 (0.83)	0.28
	Textiles/apparel/leather	10.47 (3.53)	0.47 (0.84)	3.37 (0.86)	5.06 (1.28)	
	Wood/paper/printing	-0.36 (3.62)	2.90 (0.86)	2.85 (0.78)	2.82 (0.90)	
	Petrol./chem./pharm./rubber	3.13 (2.45)	3.73 (0.98)	4.28 (1.07)	4.63 (1.21)	
	Minerals/metals	-1.20 (2.40)	5.79 (1.56)	5.57 (1.47)	5.41 (1.48)	
	Machinery/electrical	4.49 (2.42)	2.59 (0.79)	3.44 (0.89)	4.20 (1.15)	
	Transport/other manuf.	7.34 (2.16)	0.72 (0.57)	2.80 (0.77)	4.35 (1.13)	

Notes: This table replicates Table 1, but uses the trade-weighted winsorized applied-tariff measure `tariff95_w` from the Global Tariff Database. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	1.90 (0.68)	1.82 (0.45)	2.21 (0.53)	2.55 (0.62)	0.30
(B)	Consumer manuf.	1.75 (0.72)	2.18 (0.54)	2.55 (0.60)	2.89 (0.68)	0.30
	Producer manuf.	2.07 (0.76)	1.43 (0.40)	1.83 (0.49)	2.17 (0.58)	
(C)	Food, bev., tob.	1.72 (1.06)	2.92 (0.74)	3.29 (0.78)	3.62 (0.87)	0.30
	Textiles/apparel/leather	1.37 (0.71)	1.32 (0.37)	1.63 (0.38)	1.97 (0.46)	
	Wood/paper/printing	1.39 (1.05)	0.64 (0.26)	0.89 (0.32)	1.10 (0.44)	
	Petrol./chem./pharm./rubber	1.97 (0.92)	1.17 (0.31)	1.68 (0.45)	1.98 (0.56)	
	Minerals/metals	1.93 (1.01)	1.91 (0.64)	2.25 (0.68)	2.60 (0.76)	
	Machinery/electrical	2.40 (0.87)	1.62 (0.49)	1.91 (0.51)	2.27 (0.57)	
	Transport/other manuf.	2.96 (0.93)	1.27 (0.40)	1.84 (0.47)	2.37 (0.59)	

Notes: This table replicates Table 2, but uses the trade-weighted winsorized applied-tariff measure `tariff95_w` from the Global Tariff Database. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	4.33 (1.65)	3.21 (1.03)	4.10 (1.23)	4.70 (1.40)	0.26
(B)	Consumer manuf.	6.61 (2.22)	2.45 (0.78)	3.83 (1.10)	4.73 (1.35)	0.26
	Producer manuf.	1.74 (2.07)	4.17 (1.39)	4.52 (1.41)	4.76 (1.49)	
(C)	Food, bev., tob.	5.81 (3.39)	2.73 (0.84)	3.31 (0.95)	3.56 (1.03)	0.27
	Textiles/apparel/leather	11.35 (4.00)	0.73 (0.78)	3.81 (1.12)	5.57 (1.64)	
	Wood/paper/printing	-0.65 (3.46)	2.78 (1.04)	2.68 (0.86)	2.62 (0.91)	
	Petrol./chem./pharm./rubber	2.99 (2.29)	3.64 (1.23)	4.17 (1.36)	4.51 (1.49)	
	Minerals/metals	-0.92 (2.59)	5.66 (1.97)	5.49 (1.85)	5.37 (1.84)	
	Machinery/electrical	4.01 (2.37)	2.32 (0.94)	3.09 (1.09)	3.73 (1.32)	
	Transport/other manuf.	7.52 (2.78)	0.46 (0.57)	2.63 (0.91)	4.18 (1.41)	

Notes: This table replicates Table 1, but uses the winsorized MFN measure `mfn95` from the Global Tariff Database. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	1.81 (0.65)	1.81 (0.49)	2.18 (0.55)	2.51 (0.63)	0.30
(B)	Consumer manuf.	1.52 (0.64)	2.21 (0.60)	2.53 (0.64)	2.83 (0.71)	0.30
	Producer manuf.	2.11 (0.78)	1.37 (0.39)	1.78 (0.48)	2.13 (0.57)	
(C)	Food, bev., tob.	1.32 (0.99)	2.94 (0.81)	3.23 (0.82)	3.48 (0.88)	0.30
	Textiles/apparel/leather	1.32 (0.73)	1.27 (0.41)	1.57 (0.46)	1.91 (0.56)	
	Wood/paper/printing	1.60 (1.01)	0.52 (0.28)	0.81 (0.32)	1.06 (0.41)	
	Petrol./chem./pharm./rubber	1.76 (0.86)	1.20 (0.34)	1.66 (0.45)	1.93 (0.55)	
	Minerals/metals	2.20 (1.06)	1.82 (0.62)	2.22 (0.66)	2.61 (0.75)	
	Machinery/electrical	2.35 (0.98)	1.54 (0.47)	1.83 (0.51)	2.18 (0.59)	
	Transport/other manuf.	2.72 (0.92)	1.28 (0.40)	1.80 (0.46)	2.29 (0.57)	

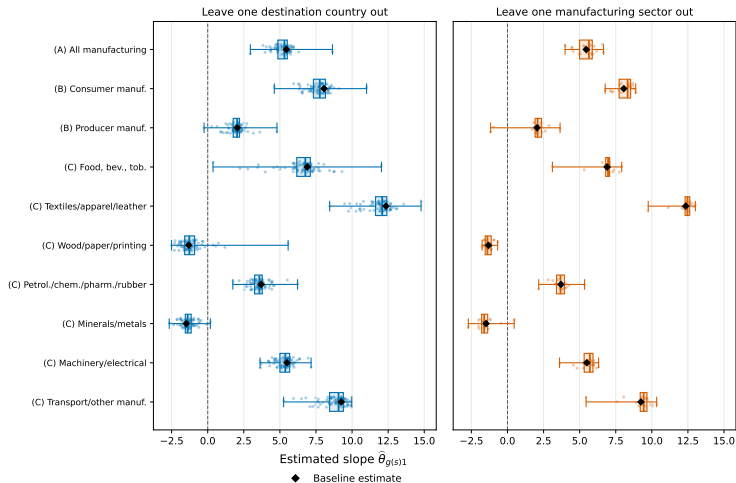
Notes: This table replicates Table 2, but uses the winsorized MFN measure `mf95` from the Global Tariff Database. Standard errors are clustered by destination country and reported in parentheses.

Spec.	Aggregate	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	7.17 (2.45)	3.32 (0.99)	4.28 (1.16)	5.34 (1.41)	0.27
(B)	Consumer manuf.	8.98 (2.78)	2.46 (0.77)	3.90 (1.02)	5.15 (1.31)	0.27
	Producer manuf.	5.24 (3.01)	4.21 (1.29)	4.51 (1.31)	5.38 (1.57)	
(C)	Food, bev., tob.	18.96 (6.75)	2.01 (0.62)	3.40 (0.73)	4.31 (0.94)	0.29
	Textiles/apparel/leather	11.49 (4.54)	0.75 (0.87)	4.00 (1.13)	5.74 (1.39)	
	Wood/paper/printing	12.51 (6.33)	1.62 (0.66)	2.71 (0.78)	4.12 (1.32)	
	Petrol./chem./pharm./rubber	6.16 (4.10)	3.67 (1.25)	4.50 (1.28)	5.30 (1.46)	
	Minerals/metals	9.55 (3.45)	5.06 (1.57)	5.15 (1.52)	6.64 (1.83)	
	Machinery/electrical	4.84 (2.74)	2.68 (0.93)	3.32 (1.11)	4.27 (1.39)	
	Transport/other manuf.	8.61 (2.89)	0.65 (0.67)	2.93 (0.91)	4.75 (1.33)	

Notes: This table replicates Table 1, but estimates $\varepsilon_{ns}(\lambda) = \theta_{s0} + \theta_{g(s)1}\lambda + \theta_{g(s)2}\lambda^2$. The derivative column is evaluated at the unweighted average across sectors of each sector's median initial domestic expenditure share. Standard errors are clustered by destination country and reported in parentheses. Each specification uses 1,510 destination-sector observations.

Spec.	Aggregate	$\widehat{\varepsilon}_{50}$	$\widehat{\varepsilon}_{25}$	$\widehat{\varepsilon}_{50}$	$\widehat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	3.07 (0.88)	2.29 (0.46)	2.67 (0.52)	3.39 (0.66)	0.31
(B)	Consumer manuf.	3.60 (1.06)	2.69 (0.53)	3.08 (0.57)	3.96 (0.74)	0.31
	Producer manuf.	2.85 (0.91)	1.81 (0.44)	2.21 (0.51)	2.81 (0.63)	
(C)	Food, bev., tob.	5.83 (1.65)	3.34 (0.70)	3.92 (0.76)	5.11 (0.97)	0.31
	Textiles/apparel/leather	1.50 (0.87)	1.48 (0.37)	1.38 (0.39)	2.46 (0.56)	
	Wood/paper/printing	3.99 (1.57)	-0.06 (0.49)	0.27 (0.46)	1.19 (0.47)	
	Petrol./chem./pharm./rubber	1.96 (1.28)	2.01 (0.54)	2.79 (0.75)	2.93 (0.75)	
	Minerals/metals	4.11 (1.35)	1.92 (0.65)	2.19 (0.63)	3.17 (0.75)	
	Machinery/electrical	3.07 (1.96)	2.27 (0.63)	2.63 (0.70)	3.12 (0.79)	
	Transport/other manuf.	3.75 (1.17)	1.76 (0.47)	2.35 (0.52)	3.11 (0.65)	

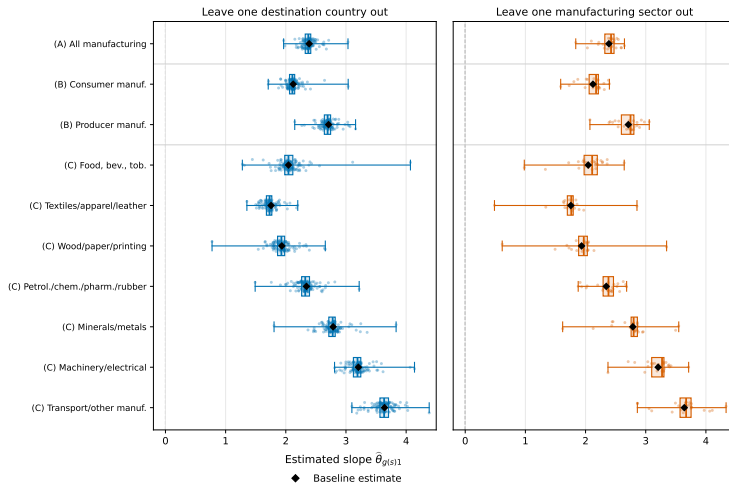
Notes: This table replicates Table 2, but estimates $\varepsilon_{ns}(\lambda) = \theta_{s0} + \theta_{g(s)1}\lambda + \theta_{g(s)2}\lambda^2$. The derivative column is evaluated at the unweighted average across sectors of each sector's median initial domestic expenditure share. Standard errors are clustered by destination country and reported in parentheses. Each specification uses 3,240 destination-sector observations.



Whiskers: minimum and maximum

Box: 25th–75th percentiles

Line: median (50th percentile)



Whiskers: minimum and maximum

Box: 25th–75th percentiles

Line: median (50th percentile)

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	3.90	4.20	5.00	5.54	0.26
(B)	Consumer manuf.	4.74	3.45	4.45	5.09	0.26
	Producer manuf.	1.42	5.13	5.41	5.61	
(C)	Food, bev., tob.	4.37	3.40	3.83	4.02	0.27
	Textiles/apparel/leather	6.63	2.68	4.48	5.51	
	Wood/paper/printing	-2.13	3.64	3.30	3.11	
	Petrol./chem./pharm./rubber	3.91	4.79	5.48	5.92	
	Minerals/metals	-1.75	7.59	7.26	7.04	
	Machinery/electrical	5.64	3.35	4.44	5.33	
	Transport/other manuf.	5.05	1.97	3.42	4.46	

Notes: This table reports constrained NLLS estimates using the OECD ICIO over 1995–2007. The estimates impose $\theta_{s0} \geq 0$ and $\theta_{g(s)1} \geq 0$ for every sector and are used in the quantification. The table reports point estimates only. Each specification uses 1,510 destination–sector observations.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	1.78	2.72	3.09	3.41	0.30
(B)	Consumer manuf.	1.76	3.10	3.48	3.82	0.30
	Producer manuf.	1.81	2.32	2.67	2.96	
(C)	Food, bev., tob.	1.93	3.96	4.38	4.75	0.30
	Textiles/apparel/leather	1.47	1.85	2.19	2.56	
	Wood/paper/printing	0.46	1.29	1.37	1.44	
	Petrol./chem./pharm./rubber	2.29	1.69	2.29	2.64	
	Minerals/metals	2.73	2.66	3.15	3.65	
	Machinery/electrical	3.14	2.33	2.72	3.19	
	Transport/other manuf.	3.56	1.95	2.63	3.26	

Notes: This table reports constrained NLLS estimates using GTAP over 2004–2014. The estimates impose $\theta_{s0} \geq 0$ and $\theta_{s0} + \theta_{g(s)1} \geq 0$ for every sector and are used in the quantification. The table reports point estimates only. Each specification uses 3,240 destination–sector observations.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	4.04	4.57	5.40	5.96	0.26
(B)	Consumer manuf.	5.17	3.77	4.85	5.55	0.26
	Producer manuf.	0.84	5.45	5.62	5.74	
(C)	Food, bev., tob.	4.18	3.74	4.15	4.33	0.27
	Textiles/apparel/leather	7.41	3.00	5.01	6.16	
	Wood/paper/printing	-2.96	3.77	3.31	3.04	
	Petrol./chem./pharm./rubber	3.15	5.21	5.77	6.12	
	Minerals/metals	-3.17	7.97	7.37	6.97	
	Machinery/electrical	8.08	3.66	5.21	6.50	
	Transport/other manuf.	5.83	2.20	3.88	5.08	

Notes: This table replicates Table 3, replacing the initial-share approximation by solving the exact demand equation forward from the observed initial domestic expenditure share and comparing its predicted terminal log home-foreign expenditure ratio with the observed ratio. The estimates use the OECD ICIO over 1995–2007. The table uses the same sample, Specifications (A)–(C), sector classifications, and constraints $\theta_{s0} \geq 0$ and $\theta_{s0} + \theta_{g(s)1} \geq 0$ as the comparison table. The table reports point estimates only. For each aggregate, $\hat{\varepsilon}_p$ is the unweighted mean across sectors of the fitted sectoral elasticities evaluated at each sector's own p th percentile of the initial domestic expenditure share. The country-cost changes and elasticity parameters are estimated jointly. The residual and R^2 remain in log home-foreign expenditure-ratio units.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	1.91	2.80	3.19	3.53	0.30
(B)	Consumer manuf.	1.97	3.17	3.59	3.97	0.30
	Producer manuf.	1.86	2.40	2.76	3.07	
(C)	Food, bev., tob.	1.91	4.01	4.43	4.80	0.30
	Textiles/apparel/leather	1.95	1.96	2.40	2.90	
	Wood/paper/printing	0.52	1.30	1.40	1.48	
	Petrol./chem./pharm./rubber	2.36	1.77	2.39	2.75	
	Minerals/metals	2.77	2.84	3.34	3.84	
	Machinery/electrical	3.69	2.49	2.94	3.50	
	Transport/other manuf.	3.81	2.14	2.87	3.55	

Notes: This table replicates Table 4, replacing the initial-share approximation by solving the exact demand equation forward from the observed initial domestic expenditure share and comparing its predicted terminal log home-foreign expenditure ratio with the observed ratio. The estimates use GTAP over 2004–2014. The table uses the same sample, Specifications (A)–(C), sector classifications, and constraints $\theta_{s0} \geq 0$ and $\theta_{s0} + \theta_{g(s)1} \geq 0$ as the comparison table. The table reports point estimates only. For each aggregate, $\hat{\varepsilon}_p$ is the unweighted mean across sectors of the fitted sectoral elasticities evaluated at each sector's own p th percentile of the initial domestic expenditure share. The country-cost changes and elasticity parameters are estimated jointly. The residual and R^2 remain in log home-foreign expenditure-ratio units.

Spec.	Aggregate	$\hat{\theta}_1$	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{50}$	$\hat{\varepsilon}_{75}$	R^2
(A)	All manufacturing	8.77 (3.81)	-0.22 (0.26)	1.42 (0.80)	2.74 (1.35)	0.23
(B)	Consumer manuf.	5.37 (2.33)	0.39 (0.33)	1.33 (0.66)	2.17 (1.00)	0.25
	Producer manuf.	10.66 (4.37)	-0.12 (0.33)	2.00 (1.01)	3.55 (1.63)	
(C)	Food, bev., tob.	4.02 (3.35)	0.67 (0.41)	0.94 (0.56)	1.22 (0.76)	0.26
	Textiles/apparel/leather	2.68 (1.98)	0.43 (0.48)	0.93 (0.61)	1.57 (0.98)	
	Wood/paper/printing	4.37 (3.18)	0.20 (0.26)	0.80 (0.53)	1.18 (0.79)	
	Petrol./chem./pharm./rubber	1.21 (1.43)	0.84 (0.54)	1.07 (0.65)	1.23 (0.77)	
	Minerals/metals	11.16 (5.57)	0.15 (0.40)	2.44 (1.32)	3.97 (2.06)	
	Machinery/electrical	8.18 (4.23)	-0.63 (0.39)	1.07 (0.61)	2.37 (1.26)	
	Transport/other manuf.	8.82 (4.51)	-0.69 (0.40)	1.14 (0.67)	2.97 (1.58)	

Notes: This table replicates Table 1, but uses OECD ICIO outcomes over 2004–2013 and MAcMap-HS6 tariff changes rather than the baseline period and tariff source. Standard errors are clustered by destination country and reported in parentheses.